How Do Web-Pretrained DINOv3 Features Extend to Martian Terrain? Appearance, Adaptation and Errors on AI4Mars

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Abstract

Foundation vision models are trained on web images, yet they are increasingly used on scenes that look nothing like the web, such as the surface of Mars. We study how the features of a frozen DINOv3 ViT-B/16 extend to terrain segmentation on AI4Mars (soil, bedrock, sand and `big_rock`) from three angles: whether the gap is one of image appearance, where in the network it pays to adapt, and where the model fails and whether it knows. A 4.6 k-parameter linear probe reaches 0.674 / 0.723 mIoU on the expert test sets M1 / M2, statistically indistinguishable from a DeepLabV3-R50 trained end to end under the same compute budget, with about 8,600 times fewer trainable parameters. Changing how the images look (histogram equalisation, CLAHE, standardisation, blur) does not move the probe, and rank-8 LoRA on early, middle or late blocks adds only 0.005–0.007 mIoU with no difference between zones. What does limit the score is label ambiguity (0.862 mIoU where three experts agree against 0.546 where only one labelled) and the rare `big_rock` class, where confidence exceeds accuracy by 0.67 even though overall calibration is good (ECE 0.029). Together, the evidence places the performance ceiling in the labels and in one rare class, not in the visual representation; the hypotheses registered beforehand were only partly decided by the data, and we say which ones.

1 Introduction

Modern computer vision increasingly relies on *foundation models*: large networks pretrained once, without labels, on hundreds of millions of images and then reused for many tasks. Models such as DINOv2 and DINOv3 [9, 8] are especially attractive because their dense features can be used *frozen*, with a very small head on top, which removes the need for large labelled sets and long training. This matters most in fields where labels are expensive, and planetary science is a clear example: every image comes from a rover on another planet, and every label is a human judgement about a scene few people have seen.

Terrain segmentation for Mars rovers is a concrete instance. A rover must tell soil from bedrock, sand and large rocks to plan a safe path, and the AI4Mars dataset [1] offers more than 35,000 NavCam images with crowd-sourced labels and a smaller expert-verified test set. The images, however, are low-contrast greyscale frames of reddish ground, very far from the colourful web photographs on which DINOv3 was pretrained. It is not obvious that features learned on the web carry over, nor what, if anything, must be changed to make them do so.

The literature has approached this in two separate ways. On one side, supervised networks trained end to end reach a mean intersection-over-union (mIoU) of about 0.67–0.70 on the hardest expert test set [3]. On the other, a few recent works use frozen or lightly adapted backbones on Mars, but with DINOv2 on a different dataset and task [6], or with a convolutional network and merged classes [7]. What is missing is a controlled look at how far a frozen DINOv3 goes on the standard AI4Mars protocol, *why* it stops where it does, and whether its confidence can be trusted on the class that matters most for safety. Without that, a practitioner cannot decide whether to spend effort on preprocessing the images, on adapting the network, or on cleaning the labels.

This paper addresses that problem. Our central question is *how do the features of a DINOv3 pretrained on web images extend to the Martian domain?* We break it into three objectives: (P1) test whether the gap is one of image *appearance*, by changing only how the photographs look; (P2) find where in the network adaptation is worth its cost, by giving the same low-rank budget to early, middle or late blocks; and (P3) characterise the *errors*, in terms of disagreement between experts and calibration. A short architectural teardown (parameters and FLOPs per block) supports these objectives but is not a question of its own. To answer them we first need to know what the earlier literature already established about the backbone, its adaptation, the Martian task and the errors that remain.

2 Literature Review

Self-supervised backbones and DINOv3. DINOv2 [9] showed that a vision transformer trained with self-distillation on curated web data yields general-purpose features that work with a linear head. DINOv3 [8] scales data and model size further and adds a regulariser, Gram anchoring, to stop dense (per-patch) features from degrading in long training, which is the property segmentation needs. Out-of-the-box use of these features has been tested on geological images: Brondolo and Beaussant [10] find that a frozen DINOv2 classifies rock images almost perfectly, that a simple head generalises better than a heavy decoder, and that LoRA helps out of distribution. They work on CT scans of

rock samples, not on rover imagery, so whether frozen features hold on Mars is open.

Adapting a frozen vision transformer. Low-rank adaptation (LoRA) [11] trains small matrices added to the linear layers and leaves the backbone untouched. ExPLoRA [12] combines it with continued self-supervised pretraining to adapt ViTs to satellite imagery, unfreezing only one or two blocks and tuning the rest with LoRA. How classic work describes transfer, early layers being general and late layers task-specific [13], suggests where to adapt, but evidence for modern ViTs under a large domain shift is mixed. The studies we found place adapters in all blocks or choose them by search; we did not find a controlled comparison of early, middle and late blocks on rover imagery, which is what P2 provides.

Martian terrain segmentation. AI4Mars provides crowd-sourced labels and expert test sets of about 322 images, built from three expert agreements (M1 to M3) [1]. Test-set sizes differ across papers (323 in [1], 322 in [4], 255 in [2]), so comparisons across them are indicative at best. End-to-end CNNs and transformers reach 0.67–0.70 mIoU on M1 [3]. Goh et al. [4] show that self-supervised pretraining helps when labels are scarce, but with accuracy on six classes. MarsUnet [7] adds parallel adapters to a frozen ResNet-50 and merges classes. MANTLE [6] puts a frozen DINOv2 ViT-S/14 on Mars, for orbital classification and boulder segmentation on its own rover data, without AI4Mars. Mars-Bench [5] benchmarks Mars tasks but excludes AI4Mars because of annotation errors.

Label ambiguity and calibration. The same annotation problem is a recurring theme: labelling terrain is ambiguous and leaves much of each image unlabelled, which makes evaluation hard and standard networks give no sign of how much to trust their predictions [14]. Müller et al. study uncertainty estimation for terrain segmentation on Curiosity data. Calibration is usually measured with the expected calibration error [15], which we adopt, but we found no report of it on AI4Mars with a foundation model, nor a per-class look at the rare `big_rock` class.

Gap. Within the 11 Mars papers we reviewed and the targeted searches above, none (i) evaluates DINOv3 on AI4Mars with four-class mIoU on M1–M3, (ii) tests whether the gap to Mars is photometric, (iii) compares adapting early, middle or late blocks under the same budget, or (iv) reports calibration on `big_rock`. This is a limited search, not an exhaustive one, and we do not claim novelty for frozen backbones or adapters in general.

3 Methodology

We turned each objective into one hypothesis, H1 to H3, each with one or two registered parts, and fixed the decision rule of every part before running anything. The experiments change a single factor at a time, so that a difference in score can be attributed to it (Table 1). The hypotheses (parts HP1, HP1b, HP2, HP3a, HP3b) were saved in a timestamped file on 02/11/2026 at 01:54, an internal record rather than an external registry. A pilot with 2,000 training images then showed us a first answer, so we saved the same rules again before launching the full-data runs, and we kept the pilot as an independent replicate: it led to the same conclusions.

Table 1: From objective to experiment.

Obj.	Factor changed	Measured	Hypothesis
P1	image appearance	probe mIoU	H1
P2	depth of LoRA (early, mid, late)	mIoU vs. probe	H2
P3	expert agreement; confidence	mIoU by agreement, ECE	H3

All experiments use the MSL NavCam subset of AI4Mars, whose greyscale images carry four classes. We trained on 15,564 images, none of which appears in a test set, and held out 500 more, chosen by acquisition time, for validation. We evaluated on the three expert-labelled sets M1, M2 and M3 of 322 images each, in which a pixel is labelled when at least 1, 2 or 3 experts agree (we read them as Swan's min1–min3 sets, an equivalence inferred from the text). Unlabelled pixels are ignored, images are resized to 512 px, and predictions are scored at label resolution, averaging the classes present in the union. We chose the epoch and every learning rate on validation, never on test.

The backbone is a DINOv3 ViT-B/16 pretrained on LVD-1689M and kept frozen. Our teardown measured 85.7 M parameters and 215 GFLOPs per 512 px image, spread evenly over 12 blocks of 7.1 M parameters and 17.8 GFLOPs each. On top of it we trained a linear probe on the final patch tokens, with 4.6 k trainable parameters, which serves as the reference for "no adaptation" in every experiment.

To test whether the gap to Mars is photometric (P1), we applied one transformation to every training and test image, retrained the probe from scratch, and compared it with the untouched images. Global histogram equalisation, CLAHE and per-image standardisation alter brightness and contrast, so a photometric gap would make them help; Gaussian blur (σ=2 px at 512 px) removes fine texture, so if the probe leans on texture it should lose accuracy, above all on sand, which is told apart from soil by texture. Each probe trained for up to 10 epochs with early stopping (patience of 2 on validation mIoU) and three seeds.

To find where adaptation pays off (P2), we added LoRA [11] of rank 8 to all linear layers, attention and MLP, of two consecutive blocks and left the rest frozen: early (blocks 1–2), middle (6–7) or late (11–12). The three zones hold the same 221 k trainable parameters, so any difference comes from where the adaptation sits and not from how much of it there is. Each run lasted two epochs (about 1,900 steps) with two seeds and a learning rate fixed from the pilot validation, and the control was the probe trained with the same recipe. An advantage of the late blocks would point to a task-level gap, and an advantage of the early blocks to a low-level, appearance-like one.

To characterise the errors (P3), we split the labelled pixels of the M1 images into high agreement, those also labelled in M3, and low agreement, those labelled in

Table 2: Hypotheses, decision rules and verdicts (full data).

Hypothesis and parts	Supported if…	Verdict
H1 the gap is not photometric		inconclusive
HP1 transforms leave the probe unchanged	every transform equivalent or lower on M1 and M2	inconclusive
HP1b blur hurts, sand most	real drop on M1 and M2, largest on sand	inconclusive
H2 late blocks help most		inconclusive
HP2 late > middle, early	late−early real on M1 and M2	inconclusive
H3 ceiling set by labels, overconfident on `big_rock`		**supported**
HP3a high agreement ≥ low by 0.05	difference ≥ 0.05, CI excludes 0	**supported**
HP3b ECE ≤ 0.05, `big_rock` gap ≥ 0.10	both, with CI above 0	**supported**

Table 3: Test mIoU (4 classes), 15,564 training images. Baselines: 1 seed, 3 epochs.

Model	Params	M1	M2
Probe, frozen DINOv3 (3 seeds)	4.6 k	0.674	0.723
DeepLabV3-R50, ImageNet	39.6 M	0.667	0.713
DeepLabV3-R50, no pretrain.	39.6 M	0.628	0.681

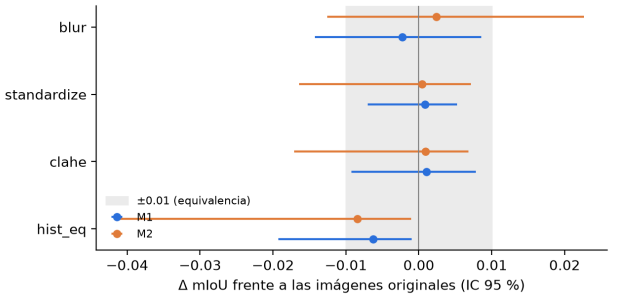


Figure 1: P1: change in probe mIoU under each appearance transform, with 95 % intervals; the grey band is the equivalence zone ±0.01.

M1 but not in M3, and compared the mIoU on each: a large gap means that part of the ceiling is set by the labels. We measured calibration on the probe's softmax confidence with the expected calibration error [15] (ECE, 15 equal-width bins) over all labelled pixels, and separately on the pixels predicted as `big_rock`, where we report mean confidence minus accuracy and the share of `big_rock` pixels the probe misses.

To place the probe against a trained network, we also trained DeepLabV3-R50 end to end on the same images, split and test sets, with and without ImageNet pretraining. The comparison is budget-matched: one seed and three epochs (about 2,900 steps) within 70 minutes per model, a budget representative of a quick deployment. Published figures use other splits, so we cite them only as context.

All differences are paired, computed on the same test images. We resampled blocks of 10 consecutive test images 2,000 times, because neighbouring frames are alike, using the same resample for every seed, and built 95 % intervals from them. We call a difference *real* when its interval excludes 0 and $|\Delta| \geq 0.01$ mIoU, *equivalent* when the interval lies inside $(-0.01, +0.01)$, and *inconclusive* otherwise; Holm's correction is applied within each claim. Table 2 states each rule and the outcome.

4 Results and Discussion

Without any adaptation, the probe matches a DeepLabV3-R50 trained end to end under the same budget (Table 3). On M1 the difference is −0.007 with a 95 % interval of $[-0.025, +0.008]$, so the data do not separate them, whereas the same network without pretraining is clearly behind (-0.046, $[-0.081, -0.016]$). The advantage of the probe is therefore one of cost: it reaches equivalent accuracy with about 8,600 times fewer trainable parameters (4.6 k against 39.6 M), and that is the point of a comparison under equal, short budgets. The ImageNet run had stopped improving by its third epoch, while the randomly initialised run was still rising, so the no-pretraining figure understates what longer training would give. Published end-to-end models reach 0.67–0.70 on M1 [3], the same range, though on a different split.

The gap is not a matter of how the images look. CLAHE and standardisation are equivalent to the original images on M1 (+0.001, intervals inside ±0.01), histogram equalisation is slightly lower (−0.006), and blurring away fine texture does not sink the probe (−0.002; Fig. 1). On M2 the intervals are wider than ±0.01, so those comparisons cannot be decided, and the data therefore do not formally confirm that every transform is neutral; still, none of them produces a gain. The texture prediction also failed: the class that loses most under blur is soil, not sand. The features appear to depend more on shape and context than on photometric statistics or on fine texture.

Adapting the backbone adds little, wherever the adaptation goes. Rank-8 LoRA gains 0.005–0.007 mIoU over the probe on M1 in every zone (Table 4; early +0.007 $[+0.001, +0.015]$, middle +0.005 $[-0.003, +0.022]$, late +0.007 $[+0.002, +0.023]$), below our 0.01 threshold, and the zones do not differ from one another (late−early: −0.000 on M1, $[-0.006, +0.015]$; +0.001 on M2). Neither the expected advantage of the late blocks nor the opposite order that a low-level gap would predict emerges from the data. An earlier layer-wise run on the same images points the same way: a 3.5 M-parameter decoder and fine-tuning of the last four blocks (28 M) added at most 0.012 on M1 and M2. Together with the appearance results, this suggests that the ceiling is not set by the representation.

Where the score does move is with the labels. On the M1 images, mIoU is 0.862 on high-agreement pixels and 0.546 on low-agreement ones, a difference of +0.315 $[+0.180, +0.413]$, so a large share of the remaining error sits exactly where the experts themselves disagree. Calibration tells a more worrying story about one class. The probe is well calibrated overall (ECE 0.029), but on

Table 4: P2: rank-8 LoRA (221 k parameters) by zone, 2 seeds, vs. the probe trained with the same recipe.

Zone	M1	M2
Probe	0.674	0.723
Early (1–2)	0.681	0.730
Middle (6–7)	0.679	0.728
Late (11–12)	0.681	0.731

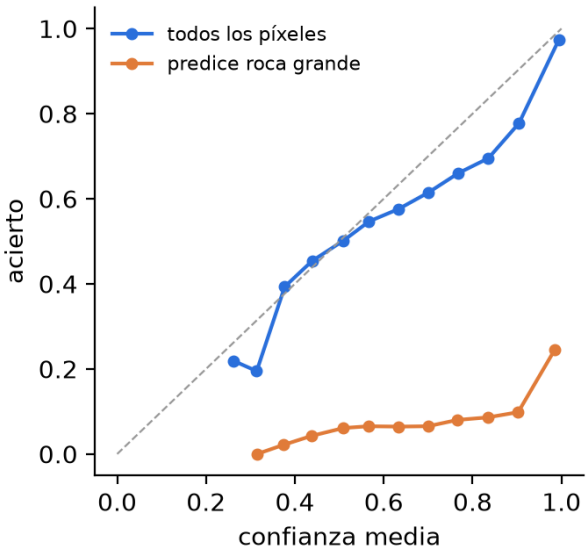


Figure 2: P3: reliability of the probe on all labelled pixels and on pixels predicted `big_rock`.

the pixels it predicts as `big_rock` its confidence exceeds its accuracy by $+0.674$ $[+0.435, +0.762]$, and 45 % of the true `big_rock` pixels are missed (Fig. 2). A single global calibration figure would hide this, and it matters for safety, because a confident "not a rock" is the dangerous error. The evaluation has a further blind spot: only 5 of the 322 M3 images contain `big_rock`, so M3 cannot rank models on that class, and its gap between heads (0.83 against 0.93) comes from false positives.

Read together, the evidence is consistent: the frozen features already work on Mars, neither preprocessing nor adaptation at any depth moves the score by more than about 0.01, and the remaining error concentrates where the labels are ambiguous and on one rare class. Two of the three hypotheses (H1 and H2) could not be decided at this sample size, which means the data neither confirm nor rule out an effect, not that the effect is zero. The teardown adds that all 12 blocks cost the same, so choosing to adapt early or late is a matter of where, not of price. In practice, effort is better spent on labels and on `big_rock` than on a heavier model.

5 Limitations, Bias and Ethics

Scope. One dataset and one backbone (DINOv3 ViT-B/16); conclusions do not extend to other rovers or models. With 322 test images and effects near 0.01 mIoU, most hypotheses cannot be decided, and M2 intervals are wider than ± 0.01.

Experimental limits. Early stopping cut most P1 probes at epoch 5, before the learning rate decayed (0.674 against 0.677 in the layer-wise run), and the three seeds share batch order, so seed variance is understated. P2 uses two seeds, two epochs and a learning rate fixed from the pilot. The baseline has one seed and three epochs, which gives a budget-matched but not a converged comparison. The pilot with 2,000 images was first presented without making clear that it differed from the full training; we corrected this and report the full-data runs.

Labels and safety. Training labels are crowd-sourced and noisy, the reason Mars-Bench leaves AI4Mars out [5]. Because `big_rock` is both rare and overconfidently predicted, these models should not be relied on for hazard detection without additional safeguards.

Data and licences. AI4Mars is public NASA data with no personal data. DINOv3 weights are used under Meta's licence; we release code and results, not weights.

Use of AI assistance. The code of the notebooks and launch scripts and drafts of the text were written with the help of an AI language model at the authors' request. The authors are responsible for the experimental design, the verification of the results and the content of this paper.

6 Conclusion

Web-pretrained DINOv3 features extend to Martian terrain with almost no adaptation: a 4.6 k-parameter probe matches an end-to-end DeepLabV3 within the statistical margin. The gap is not one of image appearance, and adapting early, middle or late blocks with the same low-rank budget adds only about 0.006 mIoU. What limits the score is the ambiguity of the labels and the rare `big_rock` class, on which the model is confidently wrong. Future work should test whether better labels or class-aware losses move this ceiling, in other domains and with larger backbones.

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